NeuroLens AI

Brain Tumor Detection Using Deep Learning

A Comparative Study of CNN, Transfer Learning, and Vision Transformer Models

Project Report
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1. Abstract

Brain tumors represent one of the most critical neurological conditions, demanding rapid and accurate diagnosis for effective clinical intervention. Manual interpretation of Magnetic Resonance Imaging (MRI) scans is time-consuming, error-prone, and heavily reliant on expert radiologist availability. This project, NeuroLens AI, proposes an automated deep learning framework for binary brain tumor classification from MRI images.

Three distinct model architectures are designed, trained, and rigorously evaluated: (1) a custom Convolutional Neural Network (CNN) baseline, (2) a Transfer Learning model leveraging the pre-trained ResNet50 backbone, and (3) a hybrid Vision Transformer (ViT) that combines ResNet50 feature extraction with transformer-based classification. Experiments conducted on a structured MRI dataset demonstrate that all three architectures achieve high classification accuracy, with the CNN baseline reaching 97.50% accuracy and 0.9928 ROC AUC, the Transfer Learning model achieving 98.25% after optimization, and the Vision Transformer attaining 98.00% with a near-perfect ROC AUC of 0.9975. The project further incorporates Grad-CAM and patch saliency explainability tools to enhance interpretability. A local web dashboard enables upload-based real-time inference. These results underscore the viability of deep learning models — particularly hybrid transformer architectures — for reliable, interpretable brain tumor detection in clinical-support settings.

2. Introduction

Brain tumors arise from abnormal cell proliferation in the brain and are among the most lethal forms of cancer worldwide. Early detection significantly improves patient outcomes, yet the diagnostic pipeline remains heavily manual. Radiologists analyze MRI scans to identify tumor presence and type, a process that is inherently subjective and limited by human capacity in high-volume clinical environments.

Deep learning has catalyzed a paradigm shift in medical image analysis over the past decade. Models trained on large annotated image datasets can learn discriminative visual features with remarkable accuracy, often matching or exceeding clinical expert performance. Three dominant architectural paradigms have emerged for image classification: Convolutional Neural Networks (CNNs), which excel at hierarchical local feature extraction; Transfer Learning models, which adapt powerful pre-trained representations to new domains; and Vision Transformers (ViTs), which capture global contextual dependencies through self-attention mechanisms.

NeuroLens AI is designed to systematically compare these three paradigms on the brain tumor detection task. The project encompasses a complete machine learning pipeline: dataset preparation, model training, evaluation, explainability analysis, and a browser-based dashboard for practical inference. By benchmarking all three architectures under controlled experimental conditions, this report provides actionable insights into the trade-offs between model complexity, computational cost, and classification performance in medical imaging.

The remainder of this report is organized as follows: Section 3 reviews relevant literature, Section 4 formalizes the problem statement, Section 5 describes the methodology, Section 6 covers explainability techniques, Section 7 details the experimental setup, Section 8 presents results and discussion, Section 9 explores applications, Section 10 outlines future scope, and Sections 11–12 conclude the report and list references.

3. Literature Review

The application of machine learning to brain MRI analysis has an extensive research history. Early approaches relied on traditional computer vision techniques such as morphological operations, support vector machines (SVMs), and handcrafted texture descriptors (GLCM, LBP) to segment and classify tumor regions. While these methods achieved moderate accuracy, they were brittle to image quality variations and required significant feature engineering expertise.

3.1 Convolutional Neural Networks in Medical Imaging

The advent of deep CNNs, exemplified by AlexNet (Krizhevsky et al., 2012) and VGGNet (Simonyan & Zisserman, 2014), transformed computer vision and subsequently medical image analysis. CNNs demonstrated an ability to automatically learn hierarchical feature representations from raw pixel data, eliminating manual feature design. In brain tumor detection, studies by Afshar et al. (2019) and Deepak & Ameer (2019) demonstrated CNN-based classifiers achieving over 94% accuracy on standard MRI datasets.

3.2 Transfer Learning for Medical Images

Transfer learning, particularly using ImageNet pre-trained models such as VGG16, ResNet50, InceptionV3, and DenseNet, has become the dominant approach for medical imaging tasks with limited labeled data. Cheng et al. (2017) demonstrated that fine-tuned ResNet architectures significantly outperform from-scratch CNNs when training data is scarce. Swati et al. (2019) applied block-wise fine-tuning of VGG19 to achieve state-of-the-art brain tumor classification results. Transfer learning models benefit from rich feature representations learned on millions of natural images, which generalize surprisingly well to MRI data.

3.3 Vision Transformers

The Vision Transformer (ViT), introduced by Dosovitskiy et al. (2020), challenged the CNN paradigm by applying the transformer self-attention mechanism — originally designed for NLP — directly to image patches. ViTs demonstrated competitive performance on ImageNet when trained on sufficient data. Subsequent hybrid models combining CNN backbones with transformer classifiers (e.g., CvT, DeiT, Swin Transformer) achieved superior results with less data, making them suitable for medical imaging. Transformer-based approaches have shown promise in capturing long-range spatial dependencies in MRI volumetric data, which CNNs struggle to model.

3.4 Explainability in Medical AI

Clinical adoption of AI diagnostic tools requires model transparency. Gradient-weighted Class Activation Mapping (Grad-CAM), introduced by Selvaraju et al. (2017), produces visual explanations by highlighting discriminative image regions contributing to a model prediction. In medical imaging, Grad-CAM has been widely adopted for tumor localization verification. For transformer models, attention rollout and patch saliency methods provide analogous explanations based on attention weight distributions across image tokens.

4. Problem Statement

Given an MRI brain scan image, the objective of NeuroLens AI is to automatically and accurately classify whether a brain tumor is present or absent. This is formulated as a binary image classification problem:

$$f(x) \rightarrow \{tumor, no\_tumor\}$$

where $x \in \mathbb{R}^{(H \times W \times C)}$ represents an input MRI image of height H, width W, and C color channels, and $f(\cdot)$ denotes the classification model mapping input images to binary class labels.

4.1 Challenges

- Intra-class variability: Tumors vary widely in size, shape, location, and MRI signal intensity across patients.
- Class imbalance: Tumor-positive cases may be underrepresented in clinical datasets.
- Limited labeled data: High-quality annotated medical datasets are expensive to construct.
- Interpretability requirements: Clinical users demand not only accurate predictions but also visual justification for model decisions.
- Generalization: Models must generalize to unseen patient scans from diverse acquisition protocols.

4.2 Success Criteria

A successful model must achieve accuracy above 95%, ROC AUC above 0.95, and produce interpretable saliency maps that highlight biologically plausible tumor regions. The system must also support practical deployment through a user-accessible web interface.

5. Methodology

5.1 Dataset and Preprocessing

The dataset is organized into three splits — train, val, and test — with class subfolders corresponding to tumor and no_tumor categories. Images are loaded using tf.keras.preprocessing.image_dataset_from_directory() with automatic label inference from folder names. All images are resized to 224×224 pixels to maintain compatibility across all three model architectures.

Data augmentation is applied during training to improve generalization and address overfitting. The augmentation pipeline includes random horizontal and vertical flips, rotation, zoom, brightness and contrast perturbation, and width/height shifts. Pixel values are normalized to the [0, 1] range using rescaling layers embedded within each model.

5.2 Model Architecture 1 — CNN Baseline

The CNN baseline (build_cnn_baseline()) is a custom architecture designed from scratch. It consists of three convolutional blocks, each comprising convolutional layers with ReLU activation, batch normalization, and max pooling. A global average pooling layer reduces spatial feature maps, followed by dropout regularization and a dense sigmoid output neuron for binary classification.

This model serves as the lower-bound reference architecture: lightweight, fast to train, and interpretable in terms of learned convolutional filters.

5.3 Model Architecture 2 — Transfer Learning (ResNet50)

The transfer learning model (build_transfer_model()) uses a ResNet50 backbone pre-trained on ImageNet. The top classification layers of ResNet50 are removed and replaced with custom dense layers, dropout, and a sigmoid output. Two training phases are implemented: (1) frozen backbone training, where only the new classification head is trained, and (2) optional fine-tuning, where deeper layers of the backbone are unfrozen and trained at a lower learning rate.

This architecture benefits from rich visual features learned on 1.2 million ImageNet images, significantly accelerating convergence on the MRI domain.

5.4 Model Architecture 3 — Hybrid Vision Transformer (ViT)

The Vision Transformer model (build_vit_classifier()) is a hybrid architecture. A frozen ResNet50 backbone acts as a patch feature extractor: the spatial feature maps output by ResNet50 are reshaped into a sequence of patch tokens. A learned positional embedding is added to encode spatial location. These patch tokens are processed through multiple transformer encoder blocks, each containing multi-head self-attention (MHSA) layers and position-wise feed-forward networks with residual connections and layer normalization. A learnable [CLS] token aggregates global context, followed by a sigmoid classification head.

5.5 Training Protocol

All models are trained using the Adam optimizer with binary crossentropy loss. Evaluation metrics tracked during training include accuracy, precision, and recall. Model checkpoints are saved to artifacts/<model>/best_weights.weights.h5 based on validation performance. Training history is serialized as .npz files and visualized through automated plotting scripts.

Training command example:

```
python src/train.py --model cnn --dataset dataset --epochs 10 --batch_size 32
```

6. Explainability Techniques

A core design requirement of NeuroLens AI is model interpretability. Clinicians must understand why a model makes a particular prediction before trusting it in diagnostic workflows. The project implements two complementary explainability approaches through src/explain.py.

6.1 Grad-CAM (CNN and Transfer Learning)

Gradient-weighted Class Activation Mapping (Grad-CAM) computes the gradient of the predicted class score with respect to the feature maps of a target convolutional layer. These gradients are globally average-pooled to obtain importance weights for each feature map channel. A weighted combination of feature maps, passed through a ReLU, produces a class activation heatmap that is upsampled and overlaid on the original MRI image.

Formally, the Grad-CAM heatmap $L^c_{\text{Grad-CAM}}$ is computed as:

$$L^c = \mathrm{ReLU}\left(\sum_k \alpha^c_k \cdot A^k \right)$$

where $\alpha^c_k = (1/Z) \sum_i \sum_j (\partial y^c / \partial A^k_{ij})$ are the global-average-pooled gradients, and A^k denotes the k-th feature map activation. This produces localized visual explanations highlighting which spatial regions of the MRI contributed most to the tumor/no_tumor prediction.

6.2 ViT Patch Saliency

For the Vision Transformer, Grad-CAM is not directly applicable due to the absence of standard convolutional layers. Instead, patch saliency maps are computed by extracting the attention weights from the final transformer encoder block's [CLS] token attention head. These attention weights, normalized across all patch positions, indicate which spatial patches the model attended to when making its classification decision.

The resulting saliency map is reshaped from the token sequence into a 2D spatial grid and upsampled to match the input image dimensions, producing an attention overlay that highlights salient tumor regions.

6.3 Output

Explainability outputs are saved to artifacts/<model>/explain/ as overlay images. These overlays are accessible through the web dashboard, allowing end users to visually inspect model decision rationale for each prediction.

7. Experimental Setup

7.1 Hardware and Software Environment

- Framework: TensorFlow / Keras
- Language: Python 3.x
- Optimizer: Adam
- Loss Function: Binary Crossentropy
- Metrics: Accuracy, Precision, Recall, F1 Score, ROC AUC
- Image Size: 224×224 pixels
- Batch Size: 32
- Epochs: 10 (baseline runs)

7.2 Dataset Split

The MRI dataset is divided into train, val, and test splits. The test split comprises 400 samples (200 tumor, 200 no_tumor) used for all final evaluations. This balanced split ensures that accuracy metrics are not confounded by class imbalance.

7.3 Evaluation Protocol

Trained model weights are loaded from saved checkpoint files and evaluated on the held-out test split using src/evaluate.py. The following metrics are computed and serialized to JSON:

- Classification Report: per-class precision, recall, F1 score
- Confusion Matrix: true positives, true negatives, false positives, false negatives
- ROC AUC: area under the receiver operating characteristic curve

Two evaluation sets are reported: real_eval_current/ (initial evaluation results) and real_eval_fixed/ (results after model fine-tuning and bug resolution), demonstrating iterative performance improvement.

7.4 Evaluation Commands

```
python src/evaluate.py --model vit --weights artifacts/vit/best_weights.weights.h5
```

8. Results and Discussion

8.1 Current Evaluation Results

Table 1 presents evaluation results for all three models on the 400-sample test split (initial evaluation run).

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
CNN	0.9750	0.9754	0.9750	0.9750	0.9928
Transfer (ResNet50)	0.8575	0.8784	0.8575	0.8555	0.9323
ViT (Hybrid)	0.9700	0.9702	0.9700	0.9700	0.9848

Table 1: Current Evaluation Metrics (400-sample test set)

8.2 Fixed/Final Evaluation Results

Table 2 presents the finalized evaluation metrics after model optimization and resolution of training issues. The Transfer Learning and ViT models show substantial improvement, both surpassing 98% accuracy.

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Transfer (ResNet50)	0.9825	0.9828	0.9825	0.9825	0.9956
ViT (Hybrid)	0.9800	0.9804	0.9800	0.9800	0.9975

Table 2: Fixed/Final Evaluation Metrics

8.3 Confusion Matrix Analysis

CNN Baseline (current run): True Negatives = 198, False Positives = 2, False Negatives = 8, True Positives = 192. The low false positive rate (2/200 = 1%) is clinically significant: falsely diagnosing a healthy patient as having a tumor causes unnecessary distress and procedures.

ViT Fixed: True Negatives = 199, False Positives = 1, False Negatives = 7, True Positives = 193. The ViT achieves near-perfect specificity (199/200 = 99.5%), the highest among all models, making it the most reliable at correctly clearing healthy patients.

8.4 Discussion

Several important observations emerge from the comparative results:

- The CNN baseline is competitive out-of-the-box (97.50% accuracy, 0.9928 ROC AUC), confirming that well-regularized custom CNNs remain highly effective for binary MRI classification.

- The Transfer Learning model initially underperforms (85.75%), likely due to domain gap or suboptimal fine-tuning configuration. After optimization, it reaches 98.25%, the highest accuracy among all models, validating the power of transfer learning when properly tuned.
- The ViT hybrid achieves the highest ROC AUC (0.9975 fixed), suggesting superior discrimination capability and the best ranking of positive versus negative cases — a critical property for clinical screening tools where threshold selection is application-dependent.
- The ViT's near-perfect specificity (false positive rate of 0.5%) makes it particularly suitable for population-level screening, where minimizing false alarms is paramount.

9. Applications

NeuroLens AI is designed with practical clinical deployment in mind. The following application domains are identified:

9.1 Clinical Decision Support

Radiologists can use NeuroLens AI as a second-opinion system, flagging MRI scans for priority review when tumor probability exceeds a defined threshold. The Grad-CAM and ViT saliency overlays provide spatially localized evidence that aligns with clinical expectation, enabling radiologists to verify or override model predictions with full transparency.

9.2 Telemedicine and Remote Diagnosis

In regions with limited access to specialist radiologists, NeuroLens AI's web dashboard enables upload-based remote screening. Healthcare workers can submit MRI images through a browser interface and receive immediate classification results with visual explanations, significantly extending diagnostic reach.

9.3 Hospital Triage Automation

High-volume hospital settings can integrate NeuroLens AI into radiology workflows to automatically pre-screen MRI scans, prioritizing suspected tumor cases for urgent radiologist review. This reduces average time-to-diagnosis for critical cases.

9.4 Research and Benchmarking

The modular architecture of NeuroLens AI makes it suitable as a benchmarking platform for evaluating new model architectures on standardized MRI classification tasks. The JSON-serialized evaluation metrics support automated performance tracking across model iterations.

9.5 Medical Education

The explainability dashboard serves as an educational tool, allowing medical students and radiology trainees to examine how deep learning models identify tumor regions in MRI scans, building intuition about both the pathology and the AI methodology.

10. Future Scope

While NeuroLens AI demonstrates strong performance, several directions for future development are identified:

10.1 Multi-class Tumor Classification

The current system performs binary tumor detection. Extending to multi-class classification — glioma, meningioma, pituitary tumor, no tumor — would substantially increase clinical utility by differentiating tumor types that require different treatment protocols.

10.2 3D Volumetric MRI Analysis

Current models process individual 2D MRI slices. Integrating 3D convolutional networks or volumetric transformers to analyze full MRI volumes would capture inter-slice context and improve detection of small or diffuse tumors not visible on individual slices.

10.3 Federated Learning

To address data privacy concerns in healthcare, federated learning frameworks can train NeuroLens AI models across multiple hospital sites without centralizing patient data. This would enable larger effective training sets while maintaining regulatory compliance (HIPAA, GDPR).

10.4 Enhanced Streamlit Dashboard

The current Streamlit dashboard requires alignment with the latest metric files and an improved upload workflow. Future development should incorporate live model comparison views, confidence interval reporting, downloadable PDF diagnostic reports, and DICOM file format support.

10.5 Dataset Expansion and Augmentation

Incorporating a larger and more diverse MRI dataset — including data from different MRI scanners, field strengths, and patient demographics — would improve model generalization. Synthetic data generation using GANs (Generative Adversarial Networks) could further augment the training distribution.

10.6 Uncertainty Quantification

Adding Bayesian inference or Monte Carlo Dropout to produce calibrated prediction uncertainty estimates would allow the system to flag low-confidence predictions for mandatory human review, improving overall reliability in safety-critical deployment.

10.7 Artifact Versioning and MLOps

Implementing consistent versioning for model checkpoints, evaluation JSON files, and training histories — integrated with a MLOps platform — would enable reproducible experiments, experiment tracking, and automated model deployment pipelines.

11. Conclusion

This report presents NeuroLens AI, a comprehensive deep learning framework for automated brain tumor detection from MRI images. Three model architectures — a custom CNN baseline, a ResNet50-based Transfer Learning model, and a hybrid Vision Transformer — are designed, trained, and benchmarked under controlled experimental conditions on a balanced 400-sample test set.

The experimental results demonstrate that all three architectures achieve clinically competitive performance. The CNN baseline provides a strong reference point at 97.50% accuracy and 0.9928 ROC AUC. The Transfer Learning model, after optimization, achieves the highest accuracy at 98.25% with 0.9956 ROC AUC. The Vision Transformer delivers the highest ROC AUC of 0.9975 and near-perfect specificity, making it the most suitable architecture for population-level screening applications.

The integration of Grad-CAM and ViT patch saliency explainability tools ensures that model predictions are interpretable and verifiable by clinical practitioners. The local web dashboard provides a practical deployment interface enabling upload-based inference without cloud dependency.

NeuroLens AI demonstrates that a modular, multi-architecture approach to medical image classification — combining established transfer learning strategies with emerging transformer paradigms — produces robust, interpretable, and clinically meaningful diagnostic AI. Future work will focus on multi-class extension, volumetric analysis, federated learning, and enhanced deployment infrastructure to transition NeuroLens AI from a research prototype toward a production-grade clinical decision support tool.

12. References

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